

Cover Letter

Candidate: Ahmet Halit Ünsal

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Position: Remote Quantitative Developer, C++ - Trading Teams EMEA

Target Organization: Tower Research Capital (Anywhere)

Job URL: <https://uk.talent.com/view?id=623057580082858400>

Dear Hiring Team & Leadership at Tower Research Capital,

I am writing to present my candidacy for the Remote Quantitative Developer, C++ - Trading Teams EMEA position at Tower Research Capital.

As the architect and developer of AURA (<https://www.auratrading.org/>), a proprietary 24/7 automated algorithmic trading and execution architecture spanning live crypto spot and equity markets, I combine deep quantitative software engineering with rigorous mathematical modeling and high-throughput system design.

My quantitative background encompasses building continuous walk-forward optimization pipelines, successive halving parameter selection, multi-regime volatility detection, and multi-layered risk management engines featuring dynamic stop-loss/take-profit triggers and PAXG defensive overlays. I architect low-latency execution pipelines using modern Python 3.11 (AsyncIO, NumPy, pandas) and C++, interfacing directly with high-frequency exchange APIs (CCXT, WebSockets) and maintaining 24/7 reliability via Linux systemd daemons and telemetry watchdogs.

In addition to quantitative finance, my track record in full-cycle product engineering (e.g., EduTrace at <https://edutrace.net>) underscores my disciplined approach to data structures, offline-first local state management, and multi-model AI workflows. I maintain rigorous test harnesses and analytical telemetry to ensure flawless runtime execution under high-stress market conditions.

I am eager to leverage this proven algorithmic engineering rigour, quantitative modeling acumen, and low-latency infrastructure design to drive outsized returns and system resilience for Tower Research Capital's quantitative strategies.

Sincerely,

Ahmet Halit Ünsal

<https://www.ahmethalitunsal.com> | <https://www.linkedin.com/in/ahmet-halit-unsal/> | <https://github.com/aunsal89>